Venkata Lakshmi Parimala Pasupuleti

Applied AI Engineer · Real Estate & PropTech · Python · LLMs · AI Agents · RAG · Data Pipelines · SQL · Forward Deployed Engineering

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SUMMARY

Applied AI Engineer with 8 years of real estate technology depth and two years building production AI systems. At Entrata - the leading PropTech SaaS platform for affordable housing and residential real estate - owned the software that powers eligibility, rent, HAP calculations, lease compliance, and HUD/LIHTC workflows for 100K+ housing units: the exact intersection of structured data, unstructured documents, and regulatory rules that defines real estate decision-making. Built **Leasel**, a production agentic AI platform built specifically for real estate: OCR-fallback document ingestion, hybrid search over lease and compliance documents, citation-grounded answers, AI-assisted root-cause investigation across settings and historical data, and a tenant self-service layer with hard eligibility boundaries. Built **AgentProbe**, an open-source LLM evaluation toolkit for production AI systems. Strong Python, SQL, REST APIs, data pipelines, and AWS; experienced with LangGraph, RAG, LLMs (Anthropic, OpenAI, Azure OpenAI), and forward deployed engineering alongside business stakeholders. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Real Estate & PropTech · Applied AI · AI Agents · LLMs · RAG · Retrieval Workflows · Python · FastAPI · Django · SQL · PostgreSQL · Data Modeling · Data Pipelines · ETL · LangGraph · LangChain · Document Intelligence · OCR · Hybrid Search · Semantic Search · Structured + Unstructured Data · AWS · S3 · RDS · Lambda · REST APIs · Integrations · HUD · LIHTC · Affordable Housing · Compliance & Regulatory Data · Stakeholder Communication · Docker · CI/CD · Anthropic · OpenAI · Azure OpenAI · Forward Deployed Engineering

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, analytical data pipelines, and AI-powered application features; surfacing business insights from raw data into APIs and reporting layers for stakeholders.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built a forward-deployed agentic solution working directly alongside business stakeholders - translated real operational workflows into an LangGraph-orchestrated AI system with RAG over regulatory documents, structured outputs, and validation guardrails.
- Deployed an MCP server connecting agentic pipeline outputs to downstream enterprise systems; engineered full ingestion, retrieval, and FastAPI serving pipeline with defined API contracts.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ housing units, multi-tenant)

- **Owned the HUD income-limit data pipeline for 7 years** - transforming federal source data into ~180,000 domain records powering rent, eligibility, HAP, and tax-credit calculations across 100K+ affordable housing units; zero compliance violations across all annual cycles; owned baseline validation, reconciliation, and rollback preparation every release.
- **Built full-stack real estate software across the compliance lifecycle** - income limits, eligibility calculations, rent schedules, lease certification documents (56 state-specific), HUD 50059 forms, annual and interim recertifications, utility allowances, and wait list workflows for LIHTC/Section 42 and HUD-regulated properties.
- **AWS S3 document migration for Affordable Housing module** - inventoried all real estate document types, mapped legacy records to migration contracts, validated metadata post-cutover; PHI-encrypted pipelines (Libsodium), 100% coverage, zero security incidents.
- Optimized 200+ SQL queries (30% performance gain) across complex multi-tenant PostgreSQL schemas for eligibility, financial, and document workflows; shipped 20+ full-stack modules across 120+ Agile sprints; mentored 5+ engineers.
- Led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction; built async distributed processing (RabbitMQ) for rent-schedule updates across thousands of tenant records.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management and affordable housing modules (Python, PHP, Java, PostgreSQL) including income-limits import (1.6M HUD/Tax Credit records across 5 years), document management, and compliance workflows; led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - Agentic AI Platform for Real Estate Compliance

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI platform combining **structured data** (settings, configurations, income limits), **unstructured documents** (lease PDFs, compliance forms, policy docs), and **historical data** (incident tickets, release notes) into a single knowledge repository - exactly the data environment Blackstone's proprietary real estate platform represents.
- **Document Intelligence**: OCR-fallback ingestion, section-aware chunking, hybrid search (TF-IDF + semantic reranking), citation grounding to exact source clauses, missing/conflicting clause detection. **Agentic Investigation**: 8-agent LangGraph pipeline producing evidence-backed root-cause hypotheses with HITL escalation. **Self-Service**: policy questions answered directly; eligibility/financial decisions always route to staff.

- 93.6% precision and 87.9% F1 across 1,240 residential lease PDFs (10 compliance rules, 5 US states + federal); CI regression harness with LLM-as-judge scoring.

AgentProbe - Open-Source LLM Evaluation Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; multi-provider LLM routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama); 5-metric eval engine (accuracy, hallucination, retrieval, completeness, compliance); CI-integrated regression harness with golden datasets - ensures production AI outputs remain reliable and accurate over time.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017